Prasad S. Gawas

PCBA Test Engineer | Electronics & Communication Engineer

Sanquelim, Goa 403505 | +91 8180941619 | prasadgawas501@gmail.com
LinkedIn Profile

Professional Summary

PCBA Test Engineer and Electronics & Communication Engineer focused on automated test-program development, board-level fault isolation, test-data analysis, and manufacturing process automation. Experienced in SMT/THT production environments, Flying Probe and multimode test workflows, Python-based engineering utilities, test sequencing, operational analytics, and hardware/software debugging. Also a published robotics researcher with interests in robotics and autonomous cyber-physical systems.

Core Competencies

PCBA Test Engineering	Flying Probe testing, multimode testing, test-program development and maintenance, test coverage, net-level fault isolation, failure analysis, and pass/fail data review.
Hardware & Production	PCB schematic interpretation, component-level debugging, signal tracing, hardware bring-up, SMT/THT processes, reflow and wave soldering, AOI and ICT workflows.
Automation & Analytics	Python, C/C++, VB.NET, LabVIEW, LabWindows/CVI, Embedded C, test sequencing, data logging, production downtime and cycle-time analysis, reporting tools.
Embedded Systems	8051, ARM Cortex and PIC architectures, peripheral interfacing, assembly-level programming, firmware debugging, PLC logic.
Test Equipment	Oscilloscope, multimeter, logic analyzer, JTAG/boundary-scan concepts, TAP-chain configuration, scripting and automation.

Professional Experience

PCBA Test Engineer **[Joining Month/Year] – Present**
Technology Department, Siemens | Verna, Goa

- Develop and maintain PCBA test programs and support SPEA Flying Probe and 3030M multimode test workflows.
- Develop and maintain PCBA test programs and support TRI ICT tester test workflows.
- Analyze test results and support board-level fault isolation, circuit failure analysis, and test-data review.
- Test and implement JTAG/Boundary scan program ausing TAP interfaces (Goepel) and analyzing JTAG failure.
- Creating FCT programs in SPEA Testers using CVI.
- Use test outcomes and production data to identify recurring faults and improve manufacturing test processes and Product coverage.

Apprentice Engineer **Sep 2025 – [End Month/Year]**
Technology Department, Siemens | Verna, Goa

- Developed software to evaluate cycle time and track downtime for DEK SMT stencil printers.
- Created bulk program generators for Ersa Three/20 and Seho reflow machines, mapping heating, cooling, convection, and CBS/Park position parameters.
- Implemented Python-based tools to analyze and report daily production-line downtime.
- Refined live-parameter comparison tools for Seho reflow and wave-soldering machines to improve parameter accuracy.
- Developed an alert-generation system to flag missing or corrupted machine programs and machine errors against daily production schedules.
- Applied model training with ETHON.AI to improve PCBA inspection efficiency on THT production lines.

Engineering Intern **Jul 2024 – Aug 2024**
Siemens | Verna, Goa

- Designed software for the Saki AOI machine to evaluate pass/fail results and diagnose defects in SMT-manufactured PCBs, contributing to a reported 30% improvement in operational efficiency without disrupting the existing system or server database.
- Collaborated with three engineering interns on automation for machine processes, stencil inventory, and shelf management.
- Developed software for a conformal-coating machine to verify PCB-palette configurations, automate coating, perform image-based final validation, and generate reports.
- Gained hands-on exposure to robotic arms, SMT/THT manufacturing processes, and collaborative-robot programming.

Selected Technical Projects

- **Production Monitoring:** Cycle-time evaluation, downtime tracking, and daily production reporting.
- **Machine Program Automation:** Bulk generation and parameter mapping for reflow programs, with alerts for missing or corrupted programs.
- **AOI Inspection Analytics:** Pass/fail analysis and defect-diagnosis software for SMT PCBs.

Education

Bachelor of Engineering – Electronics & Communication Engineering
Agnel Institute of Technology and Design (AITD), Assagao, Goa
Honours: Internet of Things (IoT)

HSSC – Science
Shri Bhumika Higher Secondary School, Parye, Goa

SSC – Goa Board
Shri Ganesh Vidhyalaya, Harvalem, Goa

Additional Skills & Interests

Robotics and mechatronics, machine learning, additive manufacturing, game development, stargazing, telescope-based sky tracking.

Languages

English (Proficient) | Konkani (Native) | Hindi (Proficient) | Marathi (Proficient)